

IGOR AUGUSTO DE MELO

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Summary

Aeronautical Engineering student at the University of São Paulo (USP) with experience in aerodynamics, propulsion systems, and aerospace vehicle design. Background in numerical simulation, flight performance modelling, and multidisciplinary engineering projects.

Experience with computational methods including CFD, Monte Carlo simulations, and optimisation techniques applied to aerospace systems. International academic experience at École Centrale de Marseille (general engineering programme).

Career Interests

Aerodynamics, Turbulence and CFD, Aeroelasticity, Thermodynamics, Flight Control Systems, Systems Engineering and Technical Project Management.

Additional interest in embedded systems and introductory cybersecurity concepts.

Education

University of São Paulo (USP) – São Carlos School of Engineering 2020 – Present
BSc in Aeronautical Engineering

École Centrale de Marseille Aug 2022 – Sep 2023
Academic Exchange Programme – General Engineering Curriculum

University of São Paulo – ICMC 2023 – Present
Emphasis in Computational Fluid Dynamics (CFD) and numerical methods (ongoing)

Engineering Experience

SEMEAR – Aerial Robotics Group (USP) Apr 2025 – Present
Project Manager – Autonomous Fixed-Wing UAV

- Leading development of an autonomous fixed-wing UAV, defining system architecture and technical requirements.
- Coordinating teams across aerodynamics, propulsion, avionics, control, and telecommunications.
- Managing system integration, validation strategy, and performance evaluation.
- Supporting flight dynamics modelling and control-oriented simulations.

Aerodynamics and Flight Dynamics Member

- Conducted aerodynamic analysis and flight performance assessments.
- Contributed to modelling and optimisation of flight performance.

TOPUS – Aerospace Projects Team

Mar 2020 – Dec 2023

- Designed and analysed solid rocket propulsion systems using thermodynamics, heat transfer, and CFD.
- Developed oxidiser flow systems (tanks, valves, injectors) with numerical modelling and performance evaluation.
- Built MATLAB-based flight simulation tools including Monte Carlo methods for trajectory dispersion and safety analysis.

- Applied systems engineering practices for requirements definition, trade-off analysis, and technical documentation.

Latin American Space Challenge (LASC)

2020 – 2022

- Contributed to structural and aerodynamic design of experimental rockets (up to 5 km apogee).
- Performed CFD simulations to improve aerodynamic efficiency.
- Achieved 3× championship titles (500 m and 1000 m categories).

Keep Flying – Aerodesign Team (USP)

2017 – 2018

- Conducted structural sizing and load analysis for high-payload aircraft.
- Performed trade-off studies between structural weight and payload capacity.

Research Experience

Unsteady Aerodynamic Load Modelling

Jul 2023 – Dec 2024

- Developed semi-empirical models for unsteady aerodynamic loads on lifting surfaces.
- Performed parameter optimisation using experimental data and CFD simulations.
- Applied optimisation and machine learning techniques (genetic algorithms, neural networks).

IRPHE – École Centrale de Marseille

Feb 2023 – Jul 2023

- Performed atmospheric simulations using the WRF model.
- Conducted uncertainty analysis for wind forecasting.
- Analysed wind turbine performance under varying atmospheric conditions.

ICMC – USP (CFD Research) – FAPESP

2023 – Dec 2024

- Participated in research activities involving turbulent flow simulation and numerical methods.
- Exposure to turbulence modelling approaches, including LES concepts.
- Development and execution of numerical routines in HPC environments.

Technical Skills

Engineering: Aerodynamics, CFD, Propulsion, Flight Mechanics, Control Systems (fundamentals), Systems Engineering, FEA

Simulation & Analysis Tools: ANSYS Fluent, CFX, OpenFOAM, XFOIL, XFLR5, WRF, Patran/Nastran (basic), Abaqus (basic)

Programming: Python, MATLAB/Simulink, SQL, basic C++ and Java, Fortran

Tools: Git, Linux, HPC, Google Cloud, CAD (SolidWorks, Fusion 360, CATIA basic), LaTeX

Languages

Portuguese (Native), English (Advanced), French (Advanced), German (Basic)